Use Case Specification

Web-Based Laboratory Inventory and Resource Management System

Scope: Junior High School Science Laboratory and Technology and Livelihood Education (TLE) Laboratory

1. Purpose

This document defines the major use cases of the Web-Based Laboratory Inventory and Resource Management System. The use cases describe how authorized users interact with the system to manage laboratory resources, inventory, borrowing transactions, equipment condition, maintenance, reports, and user access.

2. Actors

Actor	Description
System Administrator	Manages users, security, laboratory records, categories, system-wide inventory, reports, audit logs, and system configuration.
Laboratory Coordinator	Manages assigned laboratory resources, inventory, equipment, borrowing requests, returns, damage and loss, maintenance, adjustments, history, and reports.
Laboratory Staff	Performs authorized day-to-day laboratory operations such as item release, return processing, damage and loss recording, and maintenance recording.
Teacher	Views resource availability, searches inventory, submits borrowing requests, views request status and borrowing history, and confirms or facilitates returns.
Student	Interacts with laboratory resources through teachers and laboratory personnel. Students do not require individual system accounts in Version 1.0.

3. Use Case Overview

The system shall support the following major use cases. The use cases are based on the functional requirements, user roles, business rules, and core workflow defined in the SRS.

ID	Use Case	Primary Actor
UC-001	Authenticate User	All authorized users
UC-002	Manage User Account	System Administrator
UC-003	Manage Laboratory	System Administrator
UC-004	Manage Inventory	Laboratory Coordinator / authorized staff
UC-005	Manage Inventory Quantity	Laboratory Coordinator / authorized staff
UC-006	Manage Equipment and Asset	Laboratory Coordinator / authorized staff
UC-007	Search and Filter Inventory	Authorized users
UC-008	Create Borrowing Request	Teacher / authorized staff
UC-009	Approve or Reject Borrowing Request	Laboratory Coordinator / authorized staff

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UC-010	Release Borrowed Items	Laboratory Staff / authorized coordinator
UC-011	Process Returned Items	Laboratory Staff / authorized coordinator
UC-012	Record Damage, Loss, and Maintenance	Laboratory Coordinator / authorized staff
UC-013	Generate Reports	System Administrator / Laboratory Coordinator
UC-014	View Audit History	System Administrator / authorized coordinator
UC-015	View Role-Based Dashboard	Authorized users

4. Detailed Use Cases

UC-001. Authenticate User

Field	Description
Primary Actor	All authorized users
Purpose	Allow an authorized user to access protected system functions using valid credentials.
Preconditions	User account is active and valid credentials are available.
Main Flow	1. User opens the login page. 2. User enters valid credentials. 3. System validates the credentials. 4. System establishes an authenticated session. 5. System provides access based on the user's role.
Alternative / Exception	System rejects invalid credentials and does not provide access to protected resources.
Related Requirements	FR-001, FR-003, SR-001, SR-002

UC-002. Manage User Account

Field	Description
Primary Actor	System Administrator
Purpose	Create, update, activate, deactivate, assign roles, assign laboratory access, and reset user credentials when authorized.
Preconditions	System Administrator is authenticated.
Main Flow	1. Administrator opens user management. 2. Administrator selects an existing user or creates a new account. 3. Administrator enters or updates the required information. 4. Administrator assigns the appropriate role and laboratory access. 5. System validates and saves the changes. 6. System records significant user modifications in the audit trail.
Alternative / Exception	System rejects invalid data or unauthorized changes and preserves the existing account information.
Related Requirements	FR-002, FR-032, BR-011, BR-012

UC-003. Manage Laboratory

Field	Description

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Primary Actor	System Administrator
Purpose	Create and maintain configurable laboratory records.
Preconditions	System Administrator is authenticated.
Main Flow	1. Administrator opens laboratory management. 2. Administrator creates or selects a laboratory record. 3. Administrator enters laboratory information. 4. System validates the information. 5. System saves the laboratory record.
Alternative / Exception	System rejects incomplete or invalid laboratory information.
Related Requirements	5.2, 6.1, SC-001, SC-002

UC-004. Manage Inventory

Field	Description
Primary Actor	Laboratory Coordinator / authorized staff
Purpose	Register, modify, and archive inventory records while preserving transaction history.
Preconditions	User is authorized for the assigned laboratory.
Main Flow	1. User opens inventory management. 2. User creates or selects an inventory item. 3. User enters or updates inventory information. 4. System validates the data. 5. System saves the record. 6. System records significant changes in the audit trail.
Alternative / Exception	System rejects missing, invalid, duplicate, or inconsistent data.
Related Requirements	FR-004, FR-005, FR-006, FR-033, BR-001, BR-007, BR-008, BR-009, BR-010

UC-005. Manage Inventory Quantity

Field	Description
Primary Actor	Laboratory Coordinator / authorized staff
Purpose	Record stock-in, stock-out, and controlled inventory adjustments.
Preconditions	User is authorized to modify inventory quantities.
Main Flow	1. User selects an inventory item. 2. User selects stock-in, stock-out, or adjustment. 3. User enters the quantity and required transaction details. 4. System validates the transaction and reason. 5. System updates the inventory quantity. 6. System records the transaction and audit information.
Alternative / Exception	System prevents negative quantities and rejects adjustments without a valid reason.
Related Requirements	FR-008, FR-009, FR-010, FR-033, BR-003, BR-007, BR-009

UC-006. Manage Equipment and Asset

Field	Description
Primary Actor	Laboratory Coordinator / authorized staff
Purpose	Register individually tracked assets and maintain their status and condition.
Preconditions	User is authorized for the laboratory.
Main Flow	1. User registers or selects an asset.

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	2. User enters the unique Asset ID and asset information. 3. System validates the Asset ID. 4. User updates status or condition when required. 5. System saves the asset information. 6. System preserves the asset history.
Alternative / Exception	System rejects duplicate Asset IDs and invalid asset states.
Related Requirements	FR-011, FR-012, FR-013, FR-032, BR-002, BR-004, BR-005

UC-007. Search and Filter Inventory

Field	Description
Primary Actor	Authorized users
Purpose	Locate laboratory resources using available search, filtering, and sorting options.
Preconditions	User is authenticated.
Main Flow	1. User opens inventory search. 2. User enters search criteria or selects filters. 3. System retrieves matching records. 4. User reviews the results. 5. User may sort the results using supported fields.
Alternative / Exception	System displays no matching records when the criteria do not match available inventory.
Related Requirements	FR-023, FR-024, FR-025

UC-008. Create Borrowing Request

Field	Description
Primary Actor	Teacher / authorized staff
Purpose	Submit a request for laboratory resources.
Preconditions	User is authorized and active. Requested resources are known.
Main Flow	1. User checks resource availability. 2. User opens borrowing requests. 3. User enters requester, laboratory, purpose, request date, expected return date, items, quantities, and remarks. 4. System validates the request. 5. System creates the request with Pending status. 6. System records the request.
Alternative / Exception	System rejects incomplete requests. Deactivated users cannot create new borrowing requests.
Related Requirements	FR-014, FR-015, FR-033, BR-006, BR-012

UC-009. Approve or Reject Borrowing Request

Field	Description
Primary Actor	Laboratory Coordinator / authorized staff
Purpose	Review a pending request and approve or reject it.
Preconditions	A borrowing request has Pending status.
Main Flow	1. Authorized user opens the pending request. 2. User reviews the requested items, quantities, purpose, and expected return date. 3. User approves or rejects the request. 4. If rejected, the user provides a reason where

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	appropriate. 5. System updates the request status. 6. System records the action in the audit trail.
Alternative / Exception	System does not allow unauthorized users to approve or reject requests.
Related Requirements	FR-016, FR-015, FR-032, BR-006, BR-009

UC-010. Release Borrowed Items

Field	Description
Primary Actor	Laboratory Staff / authorized coordinator
Purpose	Record the physical release of approved laboratory resources.
Preconditions	Borrowing request is approved and resources are available for release.
Main Flow	1. Staff opens the approved request. 2. Staff verifies the items and quantities. 3. Staff records the release. 4. System updates the relevant inventory or asset status. 5. System changes the request status to Released. 6. System records the release transaction.
Alternative / Exception	System prevents release of unavailable, lost, or maintained assets and prevents release before approval.
Related Requirements	FR-017, FR-012, FR-033, BR-005, BR-006

UC-011. Process Returned Items

Field	Description
Primary Actor	Laboratory Staff / authorized coordinator
Purpose	Record returned resources and their condition.
Preconditions	A released borrowing transaction exists.
Main Flow	1. Staff opens the released transaction. 2. Staff records the return date and quantity returned. 3. Staff records condition, missing quantity, damaged quantity, and remarks. 4. System validates the return quantity. 5. System updates inventory or asset status. 6. System changes the request status to Returned when the transaction is complete. 7. System records the return.
Alternative / Exception	System rejects a return quantity greater than the released quantity.
Related Requirements	FR-018, FR-015, FR-033, DIR-001, DIR-002

UC-012. Record Damage, Loss, and Maintenance

Field	Description
Primary Actor	Laboratory Coordinator / authorized staff
Purpose	Record damaged or lost resources and equipment maintenance activities.
Preconditions	User is authorized to update the affected resource.
Main Flow	1. User selects the affected item or asset. 2. User records the damage, loss, or maintenance information. 3. System validates the information. 4. System updates the current condition or status when required.

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	5. System saves the record. 6. System preserves the related history.
Alternative / Exception	Lost resources are not considered available. Assets under maintenance are not available for borrowing.
Related Requirements	FR-019, FR-020, FR-021, FR-022, FR-032, BR-004, BR-005

UC-013. Generate Reports

Field	Description
Primary Actor	System Administrator / Laboratory Coordinator
Purpose	Generate inventory, borrowing, equipment, and inventory movement reports.
Preconditions	User is authenticated and has reporting permission.
Main Flow	1. User opens reports. 2. User selects the report type. 3. System retrieves the relevant records. 4. System generates the report. 5. User may print or use supported export options.
Alternative / Exception	System displays an appropriate message when no records are available for the selected report.
Related Requirements	FR-027, FR-028, FR-029, FR-030, FR-031

UC-014. View Audit History

Field	Description
Primary Actor	System Administrator / authorized coordinator
Purpose	Review significant system and inventory activities for accountability and traceability.
Preconditions	User has permission to view the applicable audit records.
Main Flow	1. User opens audit history. 2. System retrieves permitted audit records. 3. User reviews timestamp, user, action, affected entity, record ID, previous value, new value, and reason where applicable.
Alternative / Exception	Ordinary users cannot edit audit records.
Related Requirements	FR-032, SR-005, BR-009

UC-015. View Role-Based Dashboard

Field	Description
Primary Actor	Authorized users
Purpose	Provide a dashboard containing information appropriate to the user's role.
Preconditions	User is authenticated.
Main Flow	1. User opens the dashboard. 2. System identifies the user's role and laboratory access. 3. System retrieves the applicable summaries. 4. System displays the dashboard information.
Alternative / Exception	System does not display information outside the user's authorized scope.
Related Requirements	FR-026, FR-003, BR-011, BR-012

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5. Core Use Case Flow

The core workflow begins with resource registration and continues through inventory tracking, resource search, borrowing request, approval, release, use, return, condition recording, maintenance when necessary, and preservation of transaction history. This workflow supports the system definition of success in the SRS.

Step	Use Case
1	Register laboratory resource
2	Locate the resource and determine quantity, status, condition, and availability
3	Create borrowing request
4	Approve or reject request
5	Release approved resource
6	Record use and return
7	Record condition, damage, or loss
8	Perform and record maintenance when necessary
9	Preserve transaction and audit history

6. Use Case Rules

- Only authorized users may access protected functions.
- A borrowing request must be approved before items can be released.
- Only authorized users may modify inventory quantities.
- Inventory quantities shall never be negative.
- Lost resources and assets under maintenance shall not be available for borrowing.
- Completed transactions shall not be silently deleted.
- Important inventory and system changes shall create audit records.
- A Laboratory Coordinator shall only manage laboratories assigned to them unless granted system-wide privileges.
- A Teacher shall only be able to view and manage their own borrowing requests unless additional permissions are granted.

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